# 6) Suggested Fact and Dimensions

# 🏗️ COMPLETE SSAS DIMENSIONS & FACTS

## Star Schema Design for Your Data Cube

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## 🎯 INTRODUCTION

This document provides a complete **star schema design** for your SSAS cube, with all **dimensions** (the "by what") and **facts** (the "what") clearly defined.

> *"Dimensions describe the data, Facts measure the data."*

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# 📐 SECTION 1: DIMENSIONS

## Dimension 1: Date Dimension

### Description

Standard time dimension for time-based analysis and trend tracking.

### Source

Derived from `SessionDate` column.

### Attributes

| Attribute | Type | Description | Values |
|---|---|---|---|
| **DateKey** | Integer | Surrogate key (YYYYMMDD) | 20241103, 20240825 |
| **FullDate** | Date | Full date value | 2024-11-03 |
| **Year** | Integer | Year number | 2024, 2025 |
| **Quarter** | Integer | Quarter number (1-4) | 1, 2, 3, 4 |
| **QuarterName** | String | Quarter name | Q1, Q2, Q3, Q4 |

| Attribute | Type | Description | Values |
|---|---|---|---|
| **Month** | Integer | Month number (1-12) | 1-12 |
| **MonthName** | String | Month name | January, February |
| **DayOfMonth** | Integer | Day of month (1-31) | 1-31 |
| **DayOfWeek** | Integer | Day of week (1-7) | 1=Sunday, 7=Saturday |
| **DayName** | String | Day name | Monday, Tuesday |
| **WeekNumber** | Integer | Week of year (1-52) | 1-52 |
| **IsWeekend** | Boolean | Weekend flag | TRUE/FALSE |
| **IsHoliday** | Boolean | Holiday flag | TRUE/FALSE |
| **AcademicPeriod** | String | Semester/term | Fall2024, Spring2025 |

### Hierarchies

```
Year → Quarter → Month → Day
Year → Month → Day
AcademicPeriod → Year → Month
```

### Example Values

| DateKey | FullDate | Year | Quarter | MonthName | DayName | IsWeeke |
|---|---|---|---|---|---|---|
| 20241103 | 2024-11-03 | 2024 | 4 | November | Sunday | TRUE |
| 20240825 | 2024-08-25 | 2024 | 3 | August | Sunday | TRUE |
| 20250112 | 2025-01-12 | 2025 | 1 | January | Sunday | TRUE |

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## Dimension 2: Student Dimension

### Description

Captures student characteristics for segmentation and grouping.

## Source

Derived from `StudentLevel` and `Discipline` columns.

## Attributes

| Attribute | Type | Description | Values |
|---|---|---|---|
| **StudentKey** | Integer | Surrogate key | 1, 2, 3, ... |
| **StudentID** | String | Original student ID | STUDENT001 |
| **StudentLevel** | String | Academic level | Undergraduate, Graduate, PhD |
| **LevelCode** | Integer | Numeric level code | 1, 2, 3 |
| **Discipline** | String | Subject area | Computer Science, Psychology |
| **DisciplineCode** | Integer | Discipline code | 1, 2, 3, ... |
| **StudentSegment** | String | Level + Discipline combo | UG-CS, GRAD-PSY |
| **YearOfStudy** | Integer | Year in program | 1, 2, 3, 4 |
| **IsFreshman** | Boolean | First year flag | TRUE/FALSE |
| **IsSenior** | Boolean | Final year flag | TRUE/FALSE |

## Hierarchies

```
StudentLevel → Discipline → StudentID
StudentLevel → YearOfStudy
Discipline → StudentLevel
```

## Example Values

| StudentID | StudentLevel | Discipline | StudentSegment | YearOfStudy |
|---|---|---|---|---|
| S001 | Undergraduate | Computer Science | UG-CS | 2 |
| S002 | Graduate | Psychology | GRAD-PSY | 1 |

| StudentID | StudentLevel | Discipline | StudentSegment | YearOfStudy |
|---|---|---|---|---|
| S003 | PhD | Business | PHD-BUS | 3 |

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# Dimension 3: Session Dimension

## Description

Contains session-level attributes for analyzing individual sessions.

## Source

Derived from multiple source columns.

## Attributes

| Attribute | Type | Description | Values |
|---|---|---|---|
| **SessionKey** | Integer | Surrogate key | 1, 2, 3, ... |
| **SessionID** | String | Original session ID | SESSION00001 |
| **SessionLengthMin** | Float | Session duration (continuous) | 0.03-110.81 |
| **SessionLengthBin** | String | Binned session length | Short, Medium, Long |
| **LengthBinCode** | Integer | Length bin code | 1, 2, 3, 4 |
| **TotalPrompts** | Integer | Number of prompts (continuous) | 1-39 |
| **PromptCategory** | String | Binned prompt count | Low, Medium, High |
| **PromptCategoryCode** | Integer | Prompt bin code | 1, 2, 3 |
| **AI_AssistanceLevel** | Integer | AI level (continuous) | 1-5 |
| **AITier** | String | Grouped AI level | Low, Medium, High |

| Attribute | Type | Description | Values |
| --- | --- | --- | --- |
| **AITierCode** | Integer | AI tier code | 1, 2, 3 |
| **PromptDensity** | Float | Prompts per minute | TotalPrompts / SessionLengthMin |
| **IsLongSession** | Boolean | Long session flag | TRUE if > 45 min |
| **IsHighAI** | Boolean | High AI flag | TRUE if AI_Level >= 4 |
| **IsHighPrompts** | Boolean | High prompts flag | TRUE if Prompts >= 6 |

## Hierarchies

```
SessionLengthBin → AI_AssistanceLevel
AITier → PromptCategory
```

## Example Values

| SessionID | SessionLengthBin | PromptCategory | AITier | Promp |
| --- | --- | --- | --- | --- |
| SESSION00001 | Medium | High | High | 0.35 |
| SESSION00002 | Short | Low | Medium | 0.46 |
| SESSION00003 | Long | Medium | Low | 0.17 |

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# Dimension 4: Activity Dimension

## Description

Describes the type of activity/task performed during the session.

## Source

Derived from `TaskType` and `FinalOutcome` columns.

## Attributes

| Attribute | Type | Description | Values |
|---|---|---|---|
| **ActivityKey** | Integer | Surrogate key | 1, 2, 3, ... |
| **TaskType** | String | Type of task | Studying, Coding, Quiz |
| **TaskTypeCode** | Integer | Task type code | 1, 2, 3, 4, 5, 6 |
| **TaskCategory** | String | Grouped task type | Assessment, Practice, Theory |
| **TaskComplexity** | String | Task difficulty | Easy, Medium, Hard |
| **OutcomeStatus** | String | Session outcome | Completed, Failed, Withdrawn |
| **OutcomeCode** | Integer | Outcome code | 1, 2, 3, 4 |
| **IsSuccess** | Boolean | Success flag | TRUE if Completed |
| **IsFailure** | Boolean | Failure flag | TRUE if Failed |
| **IsWithdrawn** | Boolean | Withdrawn flag | TRUE if Withdrawn |

## Hierarchies

```
TaskType → OutcomeStatus
TaskCategory → TaskType
TaskComplexity → TaskType
```

## Example Values

| TaskType | TaskCategory | TaskComplexity | OutcomeStatus | IsSuccess |
|---|---|---|---|---|
| Studying | Practice | Easy | Completed | TRUE |
| Coding | Practice | Hard | Failed | FALSE |
| Quiz | Assessment | Medium | Completed | TRUE |

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# Dimension 5: Time-of-Day Dimension

## Description

Captures the time of day when sessions occurred.

## Source

Derived from `SessionDate`.

## Attributes

| Attribute | Type | Description | Values |
|---|---|---|---|
| **TimeKey** | Integer | Surrogate key | 1, 2, 3, ... |
| **Hour** | Integer | Hour of day (0-23) | 0-23 |
| **HourRange** | String | Hour range | 00:00-00:59 |
| **TimePeriod** | String | Time period | Morning, Afternoon, Evening, Night |
| **PeriodCode** | Integer | Period code | 1, 2, 3, 4 |
| **IsMorning** | Boolean | Morning flag | TRUE if 6am-11:59am |
| **IsAfternoon** | Boolean | Afternoon flag | TRUE if 12pm-4:59pm |
| **IsEvening** | Boolean | Evening flag | TRUE if 5pm-8:59pm |
| **IsNight** | Boolean | Night flag | TRUE if 9pm-5:59am |
| **TimeBin** | String | 4-hour bins | 00-03, 04-07, etc. |

## Example Values

| Hour | TimePeriod | PeriodCode | IsMorning | IsEvening |
|---|---|---|---|---|
| 9 | Morning | 1 | TRUE | FALSE |
| 14 | Afternoon | 2 | FALSE | FALSE |
| 19 | Evening | 3 | FALSE | TRUE |

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# Dimension 6: Outcome Dimension

## Description

Details the final outcome of each session.

## Source

Derived from `FinalOutcome` and `UsedAgain` columns.

## Attributes

| Attribute | Type | Description | Values |
|---|---|---|---|
| **OutcomeKey** | Integer | Surrogate key | 1, 2, 3, ... |
| **OutcomeStatus** | String | Final outcome | Completed, Failed, Withdrawn |
| **OutcomeCode** | Integer | Outcome code | 1, 2, 3, 4 |
| **IsSuccess** | Boolean | Success flag | TRUE/FALSE |
| **IsFailure** | Boolean | Failure flag | TRUE/FALSE |
| **ReturnStatus** | String | Returned? | Returned, NotReturned |
| **ReturnCode** | Integer | Return code | 1=Returned, 0=Not |
| **UsedAgain** | Boolean | Returned flag | TRUE/FALSE |
| **OutcomeGroup** | String | Grouped outcome | Successful, Unsuccessful, Incomplete |
| **RiskLevel** | String | Churn risk | Low, Medium, High |

## Example Values

| OutcomeStatus | IsSuccess | UsedAgain | OutcomeGroup | RiskLevel |
|---|---|---|---|---|
| Completed | TRUE | TRUE | Successful | Low |
| Failed | FALSE | TRUE | Unsuccessful | Medium |
| Withdrawn | FALSE | FALSE | Incomplete | High |

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# 📊 SECTION 2: FACTS

## Fact 1: Session Fact

## Description

The primary fact table recording each session with all measurable metrics.

## Grain

One row per session.

## Foreign Keys (Dimensions)

| Dimension | Foreign Key | Description |
| --- | --- | --- |
| Date Dimension | DateKey | Session date |
| Student Dimension | StudentKey | Student performing the session |
| Session Dimension | SessionKey | Session attributes |
| Activity Dimension | ActivityKey | Task type and outcome |
| Time-of-Day Dimension | TimeKey | Time of session |
| Outcome Dimension | OutcomeKey | Final outcome details |

## Measures

| Measure Name | Source | Aggregation | Description |
| --- | --- | --- | --- |
| **Session Count** | SessionKey | Count | Number of sessions |
| **Total Session Length** | SessionLengthMin | Sum | Total minutes |
| **Avg Session Length** | SessionLengthMin | Average | Average duration |
| **Max Session Length** | SessionLengthMin | Max | Longest session |
| **Min Session Length** | SessionLengthMin | Min | Shortest session |
| **Total Prompts** | TotalPrompts | Sum | Total prompts used |
| **Avg Prompts** | TotalPrompts | Average | Average prompts per session |
| **Max Prompts** | TotalPrompts | Max | Most prompts in one session |

| Measure Name | Source | Aggregation | Description |
|---|---|---|---|
| **AI Assistance Level** | AI_AssistanceLevel | Sum | Total AI level |
| **Avg AI Level** | AI_AssistanceLevel | Average | Average AI assistance |
| **Max AI Level** | AI_AssistanceLevel | Max | Maximum AI assistance |
| **Satisfaction Sum** | SatisfactionRating | Sum | Total satisfaction points |
| **Avg Satisfaction** | SatisfactionRating | Average | Average satisfaction score |
| **Min Satisfaction** | SatisfactionRating | Min | Lowest satisfaction |
| **Max Satisfaction** | SatisfactionRating | Max | Highest satisfaction |
| **Satisfaction Count** | SatisfactionRating | Count | Number of ratings |
| **Success Flag Count** | IsSuccess (derived) | Sum | Number of successful sessions |
| **Failure Flag Count** | IsFailure (derived) | Sum | Number of failed sessions |
| **Returned Flag Count** | UsedAgain | Sum | Number of returned sessions |
| **Prompt Density Sum** | PromptDensity | Sum | Total density |
| **Avg Prompt Density** | PromptDensity | Average | Average density |

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# Fact 2: Daily Aggregated Fact

## Description

Rolled-up daily summaries for faster time-based analysis.

## Grain

One row per day.

## Foreign Keys (Dimensions)

| Dimension | Foreign Key | Description |
|---|---|---|
| Date Dimension | DateKey | Date of aggregation |

## Measures

| Measure Name | Source | Aggregation | Description |
|---|---|---|---|
| **Total Sessions** | SessionKey | Count | Sessions per day |
| **Unique Students** | StudentKey | Distinct Count | Unique students per day |
| **Total Session Minutes** | SessionLengthMin | Sum | Total minutes per day |
| **Avg Session Length** | SessionLengthMin | Average | Average per day |
| **Total Prompts** | TotalPrompts | Sum | Total prompts per day |
| **Avg Prompts** | TotalPrompts | Average | Average per day |
| **Avg Satisfaction** | SatisfactionRating | Average | Satisfaction per day |
| **Success Rate** | IsSuccess | AVG(IsSuccess) | Success percentage per day |
| **Retention Rate** | UsedAgain | AVG(UsedAgain) | Return percentage per day |

| Measure Name | Source | Aggregation | Description |
|---|---|---|---|
| **AI Adoption Rate** | AI_AssistanceLevel | COUNT(AI>0) / COUNT(*) | AI usage per day |
| **Total Completed** | IsSuccess | Sum | Completed sessions per day |
| **Total Failed** | IsFailure | Sum | Failed sessions per day |
| **Total Returned** | UsedAgain | Sum | Returned students per day |

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# Fact 3: Student Summary Fact

## Description

Student-level aggregated metrics for longitudinal analysis.

## Grain

One row per student.

## Foreign Keys (Dimensions)

| Dimension | Foreign Key | Description |
|---|---|---|
| Student Dimension | StudentKey | Student being summarized |

## Measures

| Measure Name | Source | Aggregation | Descri |
|---|---|---|---|
| **Total Sessions** | SessionKey | Count | Sessio per stu |

| Measure Name | Source | Aggregation | Descri |
| --- | --- | --- | --- |
| **Total Session Minutes** | SessionLengthMin | Sum | Minute studen |
| **Avg Session Length** | SessionLengthMin | Average | Averag per stu |
| **Total Prompts** | TotalPrompts | Sum | Promp per stu |
| **Avg Prompts** | TotalPrompts | Average | Averag per stu |
| **Avg Satisfaction** | SatisfactionRating | Average | Satisfa per stu |
| **Max Satisfaction** | SatisfactionRating | Max | Highes rating p studen |
| **Min Satisfaction** | SatisfactionRating | Min | Lowes rating p studen |
| **Avg AI Level** | AI_AssistanceLevel | Average | AI usag per stu |
| **Success Rate** | IsSuccess | AVG(IsSuccess) | Succes percen |
| **Completion Rate** | FinalOutcome | COUNT(Completed)/COUNT(*) | Compl percen |
| **Retention Status** | UsedAgain | MAX(UsedAgain) | Did stu return? |
| **First Session Date** | SessionDate | Min | First sessio |
| **Last Session Date** | SessionDate | Max | Last sessio |
| **Days Active** | SessionDate | DATEDIFF(Max, Min) | Active |

| Measure Name | Source | Aggregation | Descri |
|---|---|---|---|
| **Avg Prompt Density** | PromptDensity | Average | Efficier per stu |
| **Student Lifetime Value** | Derived | Total Sessions × Avg Satisfaction | Estima value |

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# Fact 4: Monthly Trend Fact

## Description

Monthly aggregated metrics for trend analysis and forecasting.

## Grain

One row per month.

## Foreign Keys (Dimensions)

| Dimension | Foreign Key | Description |
|---|---|---|
| Date Dimension | DateKey (Month-level) | Month of aggregation |

## Measures

| Measure Name | Source | Aggregation | Descri |
|---|---|---|---|
| **Total Sessions** | SessionKey | Count | Sessio per mc |
| **Unique Students** | StudentKey | Distinct Count | Unique studen |
| **New Students** | StudentKey | First time this month | New studen |
| **Returning Students** | StudentKey | Returning this month | Return studen |

| Measure Name | Source | Aggregation | Descri |
|---|---|---|---|
| **Retention Rate** | UsedAgain | AVG(UsedAgain) | Monthl retentic |
| **Avg Session Length** | SessionLengthMin | Average | Averag duratio |
| **Avg Satisfaction** | SatisfactionRating | Average | Averag satisfa |
| **Success Rate** | IsSuccess | AVG(IsSuccess) | Monthl succes |
| **AI Adoption Rate** | AI_AssistanceLevel | COUNT(AI>0)/COUNT(*) | Monthl usage |
| **Avg AI Level** | AI_AssistanceLevel | Average | Monthl level |
| **Total Prompts** | TotalPrompts | Sum | Monthl prompt |
| **Completion Rate** | FinalOutcome | COUNT(Completed)/COUNT(*) | Monthl comple |
| **Student Health Score** | Derived | Composite score | Overal health |

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# Fact 5: AI Performance Fact

## Description

AI-specific metrics for evaluating AI effectiveness.

## Grain

One row per session, aggregated by AI level.

## Foreign Keys (Dimensions)

| Dimension | Foreign Key | Description |
|---|---|---|
| Date Dimension | DateKey | Session date |
| Student Dimension | StudentKey | Student |
| Session Dimension | SessionKey | Session attributes |

## Measures

| Measure Name | Source | Aggregation | Description |
|---|---|---|---|
| **AI Level** | AI_AssistanceLevel | As-is | AI assistance level |
| **Sessions with AI** | AI_AssistanceLevel | COUNT(AI>0) | AI usage count |
| **Sessions without AI** | AI_AssistanceLevel | COUNT(AI=0) | Non-AI count |
| **AI Success Rate** | IsSuccess | AVG(IsSuccess WHERE AI>0) | Success with AI |
| **Non-AI Success Rate** | IsSuccess | AVG(IsSuccess WHERE AI=0) | Success without AI |
| **AI Lift** | IsSuccess | AI_Success - NonAI_Success | Improvement with AI |
| **AI Satisfaction** | SatisfactionRating | AVG(Satisfaction WHERE AI>0) | Satisfaction with AI |
| **Non-AI Satisfaction** | SatisfactionRating | AVG(Satisfaction WHERE AI=0) | Satisfaction without AI |
| **AI Retention** | UsedAgain | AVG(UsedAgain WHERE AI>0) | Retention with AI |
| **Non-AI Retention** | UsedAgain | AVG(UsedAgain WHERE AI=0) | Retention without AI |
| **AI Completion Rate** | IsSuccess | AVG(IsSuccess WHERE AI>0) | Completion with AI |
| **AI Optimal Level** | Derived | AI_Level with highest success | Best AI level |

# Fact 6: Task Performance Fact

## Description

Task-specific performance metrics.

## Grain

One row per task type per session.

## Foreign Keys (Dimensions)

| Dimension | Foreign Key | Description |
|---|---|---|
| Date Dimension | DateKey | Session date |
| Activity Dimension | ActivityKey | Task type details |
| Student Dimension | StudentKey | Student |

## Measures

| Measure Name | Source | Aggregation | Descri |
|---|---|---|---|
| **Task Count** | ActivityKey | Count | Sessio per tas |
| **Task Success Rate** | IsSuccess | AVG(IsSuccess) | Succes per tas |
| **Task Avg Satisfaction** | SatisfactionRating | Average | Satisfa per tas |
| **Task Avg Session Length** | SessionLengthMin | Average | Duratic per tas |
| **Task Avg Prompts** | TotalPrompts | Average | Promp per tas |
| **Task Avg AI Level** | AI_AssistanceLevel | Average | AI per |

| Measure Name | Source | Aggregation | Descri |
|---|---|---|---|
| **Task Retention** | UsedAgain | AVG(UsedAgain) | Retent per tas |
| **Task Completion Rate** | FinalOutcome | COUNT(Completed)/COUNT(*) | Compl per tas |

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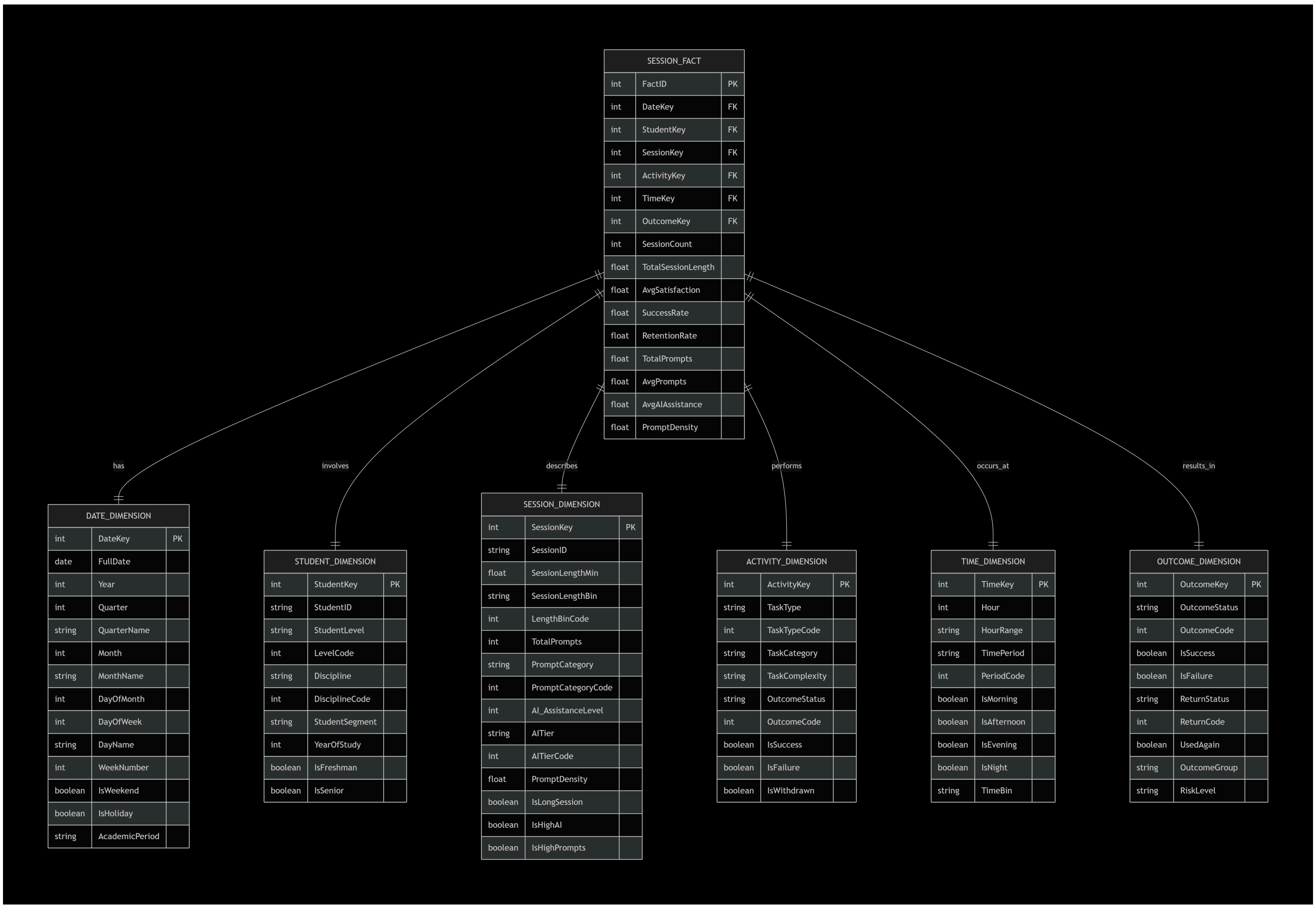
# 📊 SECTION 3: STAR SCHEMA DIAGRAM

## 📊 SSAS STAR SCHEMA - MERMAID DIAGRAM

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# 📋 SECTION 4: ADDITIONAL AGGREGATED FACTS

## Fact 7: Student Monthly Fact

| Measure | Aggregation | Description |
| --- | --- | --- |
| Sessions per Month | COUNT | Monthly activity level |
| Minutes per Month | SUM | Time invested |
| Avg Satisfaction per Month | AVG | Monthly satisfaction trend |
| Success Rate per Month | AVG(IsSuccess) | Monthly success trend |
| AI Usage per Month | AVG(AI_Level > 0) | Monthly AI adoption |

## Fact 8: AI Level Performance Fact

| Measure | Aggregation | Description |
| --- | --- | --- |
| Sessions by AI Level | COUNT | Distribution of AI levels |
| Success by AI Level | AVG(IsSuccess) | Effectiveness per level |
| Satisfaction by AI Level | AVG(Satisfaction) | Satisfaction per level |
| Retention by AI Level | AVG(UsedAgain) | Retention per level |

## Fact 9: Discipline Performance Fact

| Measure | Aggregation | Description |
| --- | --- | --- |
| Success by Discipline | AVG(IsSuccess) | Performance per subject |
| Satisfaction by Discipline | AVG(Satisfaction) | Satisfaction per subject |
| Retention by Discipline | AVG(UsedAgain) | Retention per subject |
| Avg Session Length by Discipline | AVG(SessionLengthMin) | Engagement per subject |

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# 🎯 SECTION 5: IMPLEMENTATION SUMMARY

## Dimensions Summary

| # | Dimension Name | Key Attributes | Hierarchies |
|---|---|---|---|
| 1 | Date Dimension | 14 attributes | Year→Quarter→Month→Day |
| 2 | Student Dimension | 9 attributes | Level→Discipline→Student |
| 3 | Session Dimension | 14 attributes | Length→AI Level |
| 4 | Activity Dimension | 8 attributes | Task→Outcome |
| 5 | Time-of-Day Dimension | 9 attributes | Period→Hour |
| 6 | Outcome Dimension | 9 attributes | Outcome→Risk |

## Facts Summary

| # | Fact Name | Grain | Measures |
|---|---|---|---|
| 1 | Session Fact | Per Session | 19 measures |
| 2 | Daily Aggregated Fact | Per Day | 13 measures |
| 3 | Student Summary Fact | Per Student | 14 measures |
| 4 | Monthly Trend Fact | Per Month | 13 measures |
| 5 | AI Performance Fact | Per AI Level | 12 measures |
| 6 | Task Performance Fact | Per Task Type | 8 measures |
| 7 | Student Monthly Fact | Per Student-Month | 5 measures |
| 8 | AI Level Performance Fact | Per AI Level | 4 measures |
| 9 | Discipline Performance Fact | Per Discipline | 4 measures |

## Total Count

| Element | Count |
|---|---|
| **Dimensions** | 6 |
| **Fact Tables** | 9 |
| **Total Attributes** | ~60 |
| **Total Measures** | ~90 |
| **Total Hierarchies** | 10+ |

**Bottom Line:** This complete star schema provides comprehensive analytical capabilities, supporting all reporting and analysis requirements from basic metrics to complex multidimensional analysis. 🎯